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Price Discrimination: An Introduction



Outline

- An Introduction to Price Discrimination
- Types of Price Discrimination
- First Degree Price Discrimination or Perfect Price Discrimination
- Third Degree Price Discrimination
- Some Final Remarks

A Step Back: Uniform Pricing I

- A generic monopolist might have a profit function

$$\pi(p) = (p - c) Q(p) \quad (1)$$

where c is the constant marginal cost of production faced by the monopolist and $Q(p) = a - bp$ is the usual linear demand.

- The optimal monopoly price p^m imposed to consumers is derived from (1) as follows:
 - The monopolist wants to choose the price of its good in such a way that its profit is maximized, in other words he wishes to solve a maximization problem

$$\max_p \pi(p)$$

- To do so it adopts a FOC

$$a - bp + (p - c)(-b) = 0$$

A Step Back: Uniform Pricing II

3. To then isolate p yielding

$$p^m = \frac{a}{2b} + \frac{c}{2} \quad (2)$$

Equation (2) is the price maximizing the profit function of the monopolist, therefore it is the price that will be imposed to consumers.

- Eventually the quantity of good sold by the monopolist will be obtained by plugging (2) into $Q(p) = a - bp$ yielding

$$Q(p^m) = \frac{a}{2} - \frac{bc}{2} \quad (3)$$

and the associated profit by plugging (2) and (3) into (1)

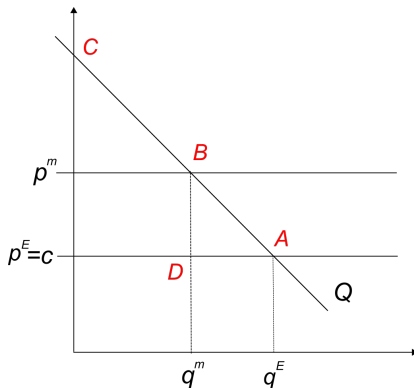
$$\pi(p^m) = \frac{a^2}{4b} - \frac{ac}{2} + \frac{c^2b}{4} \quad (4)$$

with $\frac{a}{b} > c$ to ensure (3) and (4) positive.

Uniform Pricing and Consumers' Welfare I

- The total surplus is equal to the sum of the consumer surplus and the producer surplus (or profit)
- Uniform pricing leaves some surplus to the consumers.
- Price discrimination, i.e. imposing different prices to different consumers, is aimed at turning as much consumers' surplus as possible into producer surplus.

Uniform Pricing and Consumers' Welfare II



where $p^m CBp^m$ is the consumers' surplus, $cBp^m c$ is the producer's surplus (or profit) and ADB is the monopoly dead-weight welfare loss.

Some Examples of Price Discrimination

- As noted the goal of the producer when price discriminating is exactly that of taking over the consumers' surplus.
- Examples of the same economic good sold at different prices abound:
 - An amusement park may be selling an entrance ticket for a flat fee and single rides tickets;
 - The same economic good may be sold at two different prices to the same consumer, in the case for instance of quantity discounts;
 - A doctor may charge a rich patient more than a poor one, or an insured one more than an uninsured one.

First Degree Price Discrimination

- First degree price discrimination is also known as perfect price discrimination.
- It occurs when a single consumer (or equivalently a number of identical consumers) has unit demand.
- Suppose that each consumer i has a gross utility function $w_i(q)$ indicating his evaluation for a given quantity q of a good, then the monopolist by charging a well designed *tariff* to each different consumer is able to extract the whole consumers' surplus or even the entire social welfare.

Second Degree Price Discrimination

- To practice first degree price discrimination, the monopolist needs information on consumers' tastes, which are often difficult to obtain.
- If the lack of information, or some legal reasons, force the monopolist to charge the same tariff to each consumer, it might be the case that making consumers self-selecting between offers allow him to earn a higher profit.
- Suppose that there are two types of consumers, ω_1 with probability λ and ω_2 with probability $1 - \lambda$, then the monopolist will design two contracts (q_1, t_1) and (q_2, t_2) in such a way that a consumer of type ω_1 prefers the first while one of type ω_2 prefers the second.

Third Degree Price Discrimination

- Suppose that a monopolist produces a single product and that he is able to divide the demand into m groups on the basis of some exogenous information.
- This information might be: age, sex, occupation location or first-time vs. second-time buyer.
- These m groups have m distinct demand curves for the product which are known by the monopolist.
- Hence, the monopolist is able to charge a different price for each group.

First Degree Price Discrimination: Introduction

- Suppose that in a market there is only one consumer i whose gross utility function is observable, then by using an appropriate *tariff* the monopolist not only can extract all the consumer's surplus, but it can even exploit the entire social surplus.
- A *tariff* is simply a pricing schedule which might be linear $T(q) = pq$ or affine $T(q) = F + pq$ where F is a fixed fee.
- In particular the monopoly's strategy consists of observing the consumer's product evaluation $w_i(q)$ and to design a take-it or leave-it offer, consisting of a quantity and an associated tariff maximizing his profit.

Linear Tariff Design I

- The monopolist knows that by imposing a tariff $T(q) = pq$ the consumer i has surplus of

$$S_i(q) = w_i(q) - T(q)$$

- Therefore he knows that i consumes only when

$$S_i(q) = w_i(q) - T(q) \geq 0$$

- The monopolist then can increase $T(q)$ up to $w_i(q)$, $T(q) = w_i(q)$, without making the consumer refrain from buying and consequently he seeks a q such that its profit is maximized.
- In practice the take-it or leave-it offer, consisting of a q maximizing the monopolist's profit and the associated $T(q)$, is found as follows:

Linear Tariff Design II

- The monopolist wishes to solve the following profit maximization problem:

$$\max_q \pi(q) = T(q) - cq$$

s.t. $T(q) = w_i(q)$ where c is the usual marginal cost.

- As usual we might show that a solution to this problem exists and that it might be found by the FOC:

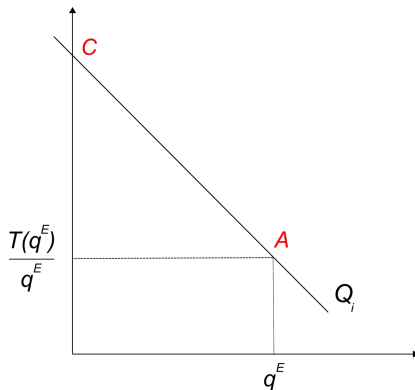
$$w'_i(q) - c = 0$$

obtaining the efficient quantity q^E such that

$$w'_i(q^E) = c$$

- This leads to the monopolist offering the consumer a take-it or leave-it offer q^E and $T(q^E) = w_i(q^E)$.

Linear Tariff Design III



This is the best possible situation for the monopolist as he managed to capture all the social welfare $A \frac{T(q^E)}{q^E} BA$ and it is therefore Pareto Efficient. Still the distribution is unfair for it leaves the consumer with zero surplus.

Example of Linear Tariff Design I

Assume a market with a monopolist m facing a constant marginal cost $c = 5$ making a take-it or leave-it offer to one consumer whose willingness to pay is

$$w_i(q) = 40\sqrt{q}$$

Then following the steps highlighted above we proceed as follows.

1. m fixes $T(q) = w_i(q) = 40\sqrt{q}$
2. m solves the maximization problem

$$\max_q \pi(q) = T(q) - cq$$

s.t. $T(q) = 40\sqrt{q}$ and $c = 5$ through the FOC

$$\frac{40}{2\sqrt{q}} - 5 = 0$$

Example of Linear Tariff Design II

3. This yields

$$q^E = 16$$

and

$$T(q^E) = 40\sqrt{16} = 160$$

4. Eventually,

$$\pi(q^E) = T(q^E) - cq^E = 80 = w_i(q^E)$$

and

$$S_i(q^E) = w_i(q^E) - T(q^E) = 0$$

Affine Tariff Design I

- Another feasible approach to first degree price discrimination is to design an affine two-part tariff rather than a linear one

$$T(q) = F + pq$$

in such a way that the consumer buys again q^E while paying $w_i(q^E)$.

- In other words, F might be considered as an "entry" fee to pay before buying anything from the monopolist.

Affine Tariff Design II

- Again the monopolist knows that the consumer will consume only in case of a positive surplus:

$$S_i(q) = w_i(q) - T(q) = w_i(q) - (F + pq) \geq 0$$

- The monopolist might therefore set F as high as $w_i(q^E) - cq^E$ and p as low as c such that

$$S_i(q) = w_i(q) - (w_i(q^E) - cq^E + cq) \geq 0$$

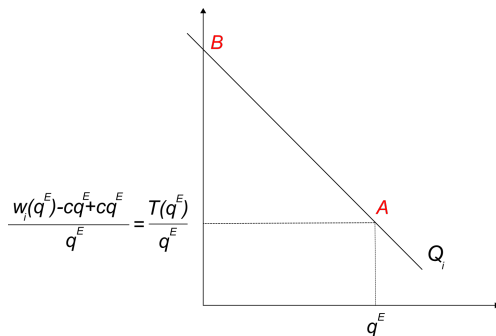
- In this way, to avoid a negative surplus, the consumer is forced to buy at least q^E at $T(q^E)$ so that

$$S_i(q^E) = w_i(q^E) - (w_i(q^E) - cq^E + cq^E) = 0$$

- Being q^E the maximum quantity the monopolist is willing to produce, he is able again to exploit the entire social welfare as

$$\pi(q^E) = w_i(q^E)$$

Affine Tariff Design III



This is one more time the best possible situation for the monopolist as he is able to get the entire social welfare $A \frac{T(q^E)}{q^E} B A$ and it is therefore Pareto Efficient. Again the distribution is unfair for it leaves the consumer with zero surplus.

Example of Affine Tariff Design I

Suppose that $w_i(q) = 40\sqrt{q}$ and $c = 5$ and consider the two-part tariff with $p = c = 5$ and $F = w_i(q^E) - cq^E = 40\sqrt{q^E} - 5q^E$.

Then following the steps highlighted above we proceed as follows.

1. i pick q such that

$$S_i(q) = w_i(q) - (F + cq) = 0$$

2. To do so it is enough for him to observe that

$$S_i(q) = 40\sqrt{q} - (40\sqrt{q^E} - 5q^E + 5q) = 40\sqrt{q} - (80 + 5q)$$

for $q^E = 16$ from the previous example.

Example of Affine Tariff Design II

3. Therefore q is chosen such that

$$40\sqrt{q} - (80 + 5q) = 0$$

which is exactly

$$q = q^E = 16$$

It is obvious by construction that the surplus of the consumer gets to zero

$$S_i(q^E) = 40\sqrt{16} - (80 + 5 \cdot 16) = 0$$

as it was entirely absorbed by the profit of the monopolist:

$$\pi(q^E) = F(q^E) + pq^E - cq^E = w_i(q^E) - cq^E$$

which is equivalent to

$$\pi(q^E) = 80$$

An Exercise on First Degree Price Discrimination

Suppose there are two consumers, with willingness to spend or (gross utility function) $w_1(q) = 40\sqrt{q}$ and $w_2(q) = 25q - \frac{1}{2}q^2$ and $c = 5$. Following the steps listed above show that the linear tariffs are $(q_1^E, T(q_1^E)) = (16, 160)$ and $(q_2^E, T(q_2^E)) = (20, 300)$ while the affine tariffs are such that $p_1 = 5$, $F_1 = 80$ and $p_2 = 5$, $F_2 = 200$.



Third Degree Price Discrimination: Introduction

- Third degree price discrimination consists in partitioning consumers in groups (for instance, old consumers and non-old consumers, or males and females, or employed and unemployed people), and then charging different prices to different groups.
- Optimal pricing implies that the monopolist should charge more in markets with the lower elasticity of demand.
- This yields a higher profit with respect to uniform pricing, in which case the monopolist is constrained to charge the same price to both groups.

Example of Third Degree Price Discrimination I

Suppose that in a market with n consumers and only one producer facing marginal cost $c = 1$, the total market demand can be partitioned in two:

$$Q_1(p) = \frac{2000}{p^3} \text{ and } Q_2(p) = \frac{2000}{p^2}.$$

If price discrimination is feasible, there are two different markets such that the profit of the monopolist is

$$\pi_1(p_1) = (p_1 - 1)Q_1(p_1) = (p_1 - 1)\frac{2000}{p_1^3}$$

in the first market and

$$\pi_2(p_2) = (p_2 - 1)Q_2(p_2) = (p_2 - 1)\frac{2000}{p_2^2}$$

in the second.

Example of Third Degree Price Discrimination II

Then we can find the optimal price, quantity and profit in the first market as follows:

1. As usual the monopolist wishes to solve the typical profit maximization problem

$$\max_{p_1} \pi_1(p_1)$$

2. Then a unique solution exists which might be found by the FOC

$$\frac{2000}{p_1^3} - \frac{3p_1^2 2000}{p_1^6} (p_1 - 1) = 0$$

3. This yields $p_1^* = \frac{3}{2}$, $Q_1(p_1^*) = \frac{2000}{27/8} \cong 592.6$ and $\pi_1(p_1^*) = \left(\frac{3}{2} - 1\right) \frac{2000}{27/8} \cong 296.3$

Example of Third Degree Price Discrimination III

We proceed identically for the second market:

1. The monopolist wants to solve the problem

$$\max_{p_2} \pi_2(p_2)$$

2. Then a unique solution exists which might be found by the FOC

$$\frac{2000}{p_2^2} - \frac{2p_2 2000}{p_2^4} (p_2 - 1) = 0$$

3. This yields $p_2^* = 2$, $Q_2(p_2^*) = \frac{2000}{4} = 500$ and $\pi_2(p_2^*) = (2 - 1) \frac{2000}{4} = 500$

Example of Third Degree Price Discrimination IV

As we expected $p_1^* = \frac{3}{2} < 2 = p_2^*$ and the absolute value of the elasticity $|E_1(p_1)|$ of market 1 is greater than the elasticity $|E_2(p_2)|$ of market 2. Recall that

$$|E_1(p_1)| = \left| \frac{dQ_1(p_1)}{dp_1} \frac{p_1}{Q_1(p_1)} \right| = \left| -\frac{3p_1^2 2000}{p_1^6} \frac{p_1}{2000/p_1^3} \right| = 3$$

and

$$|E_2(p_2)| = \left| \frac{dQ_2(p_2)}{dp_2} \frac{p_2}{Q_2(p_2)} \right| = \left| -\frac{2p_2 2000}{p_2^4} \frac{p_2}{2000/p_2^2} \right| = 2$$

We eventually note that $\pi_1(p_1^*) < \pi_2(p_2^*)$.

Example of Third Degree Price Discrimination V

What if price discrimination is not feasible?

Let's take the usual steps considering a unique price p and a unique market $Q(p) = Q_1(p) + Q_2(p) = \frac{2000}{p^2} + \frac{2000}{p^3} = \frac{2000(p+1)}{p^3}$

1. The monopolist looks for the solution of

$$\max_p \pi(p) = (p - 1) Q(p)$$

2. A unique solution is recovered by the FOC

$$2000 \frac{2p^4 - 3p^2(p^2 - 1)}{p^6} = 0$$

3. It yields $p^* = \sqrt{3}$, $Q(p^*) = \frac{2000}{3\sqrt{3}} \cong 1050.79$ and $\pi(p^*) = \frac{4000}{3\sqrt{3}} = 770$

Welfare Analysis of Third Degree Price Discrimination I

- By the previous example it is immediate to see that the monopolist gains from price discrimination as

$$(\pi(p^*) = 770) < (796.3 = 296.3 + 500 = \pi_1(p_1^*) + \pi_2(p_2^*))$$

- Similarly

$$\left(p_1^* = \frac{3}{2}\right) < \left(p^* = \sqrt{3}\right) < (p_2^* = 2)$$

meaning that the group with high abs. val. elasticity gains from price discrimination at the expenses of the group with low elasticity.

- This is always so, except for the case where the highly abs. val. elastic group is excluded from the aggregate market.

Welfare Analysis of Third Degree Price Discrimination II

- For these reasons the the welfare effects of third degree price discrimination are ambiguous:
 - A policy maker should indeed weigh the losses of consumers in low abs. val. elasticity markets against the gains of those in high abs. val. elasticity markets and of the producer.
 - The elimination of price discrimination may be particularly dangerous if it leads to the closure of markets.

Final Remarks

- Second degree price discrimination was not treated as partial information concerning tastes of consumers require other tools from microeconomic theory which exceed the scope of this lecture.
- Next lectures will analyze price discrimination in Hotelling's and in Salop's spatial competition models.

References

Menicucci, Domenico. 2016. "Basic Monopoly Theory". Unpublished Class Notes.

Luporini, Annalisa. 2024. "Price Discrimination". Unpublished Class Notes.

Tirole, Jean. 1994. "The Theory of Industrial Organization". The MIT Press: Cambridge, Massachusetts.